

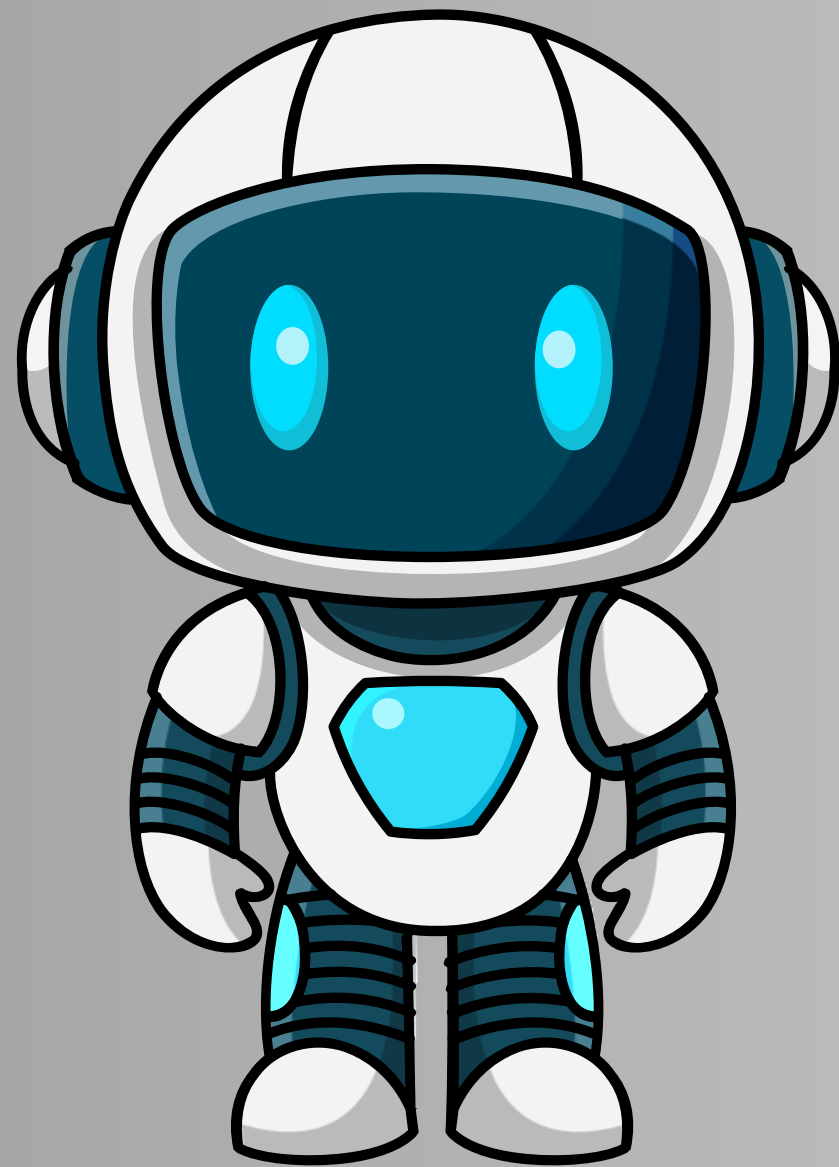
GENERATIVE AI

BY

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PROJECT OVERVIEW



Project Goal

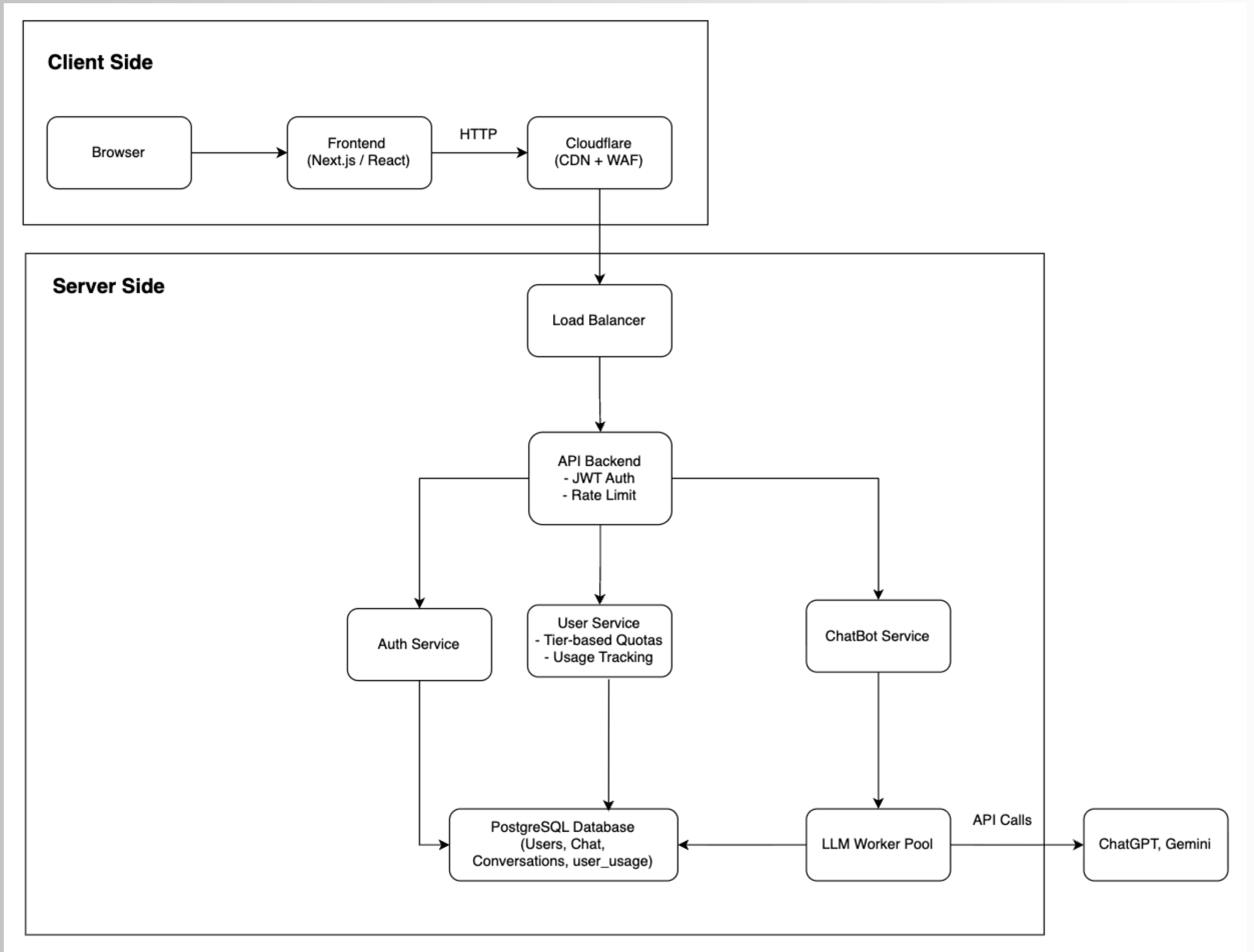
Build a production-grade Generative AI platform that delivers secure, scalable, and cost-effective LLM inference at enterprise scale.

What is AgathaAI?

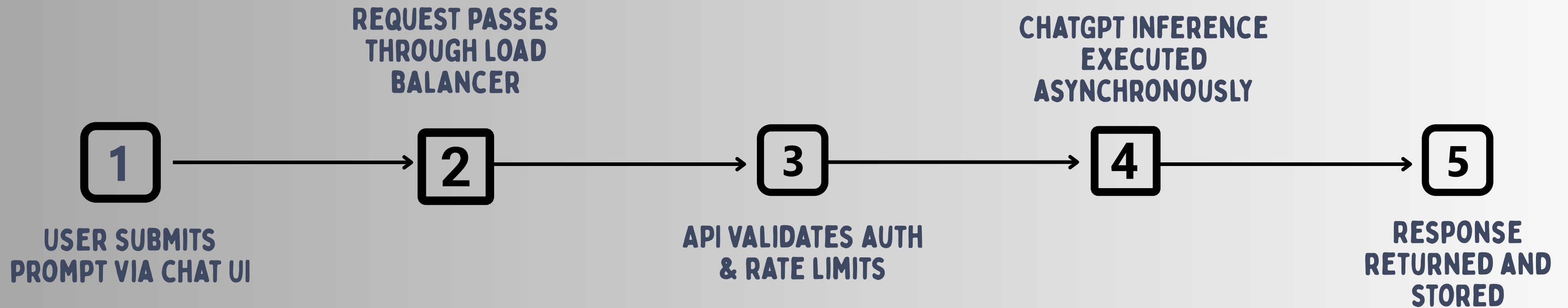
A cloud native ChatGPT-like platform with:

- Multi-provider LLM integration (ChatGPT + Gemini)
- Asynchronous inference processing
- Enterprise-grade security and scalability

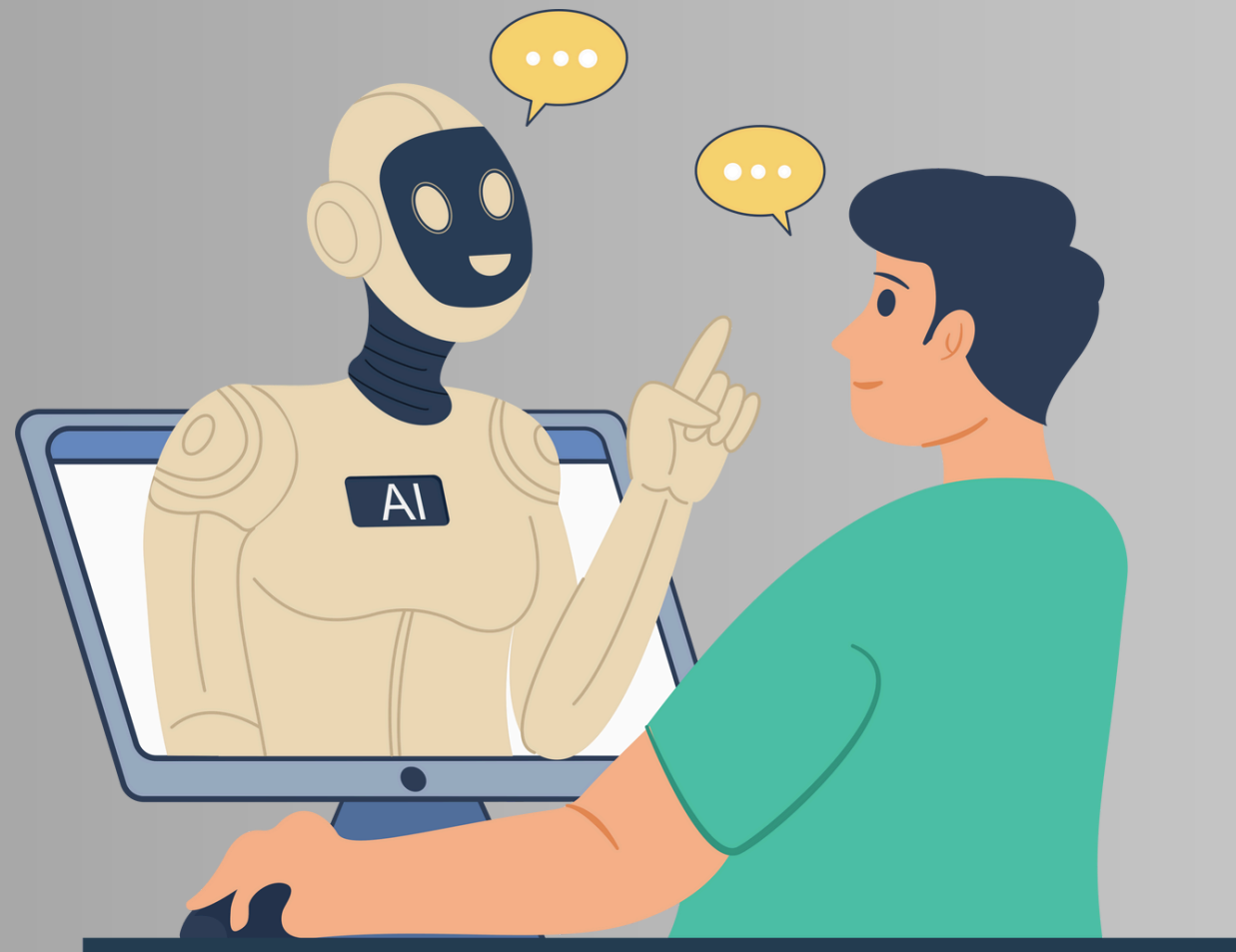
SYSTEM DIAGRAM



HOW THE SYSTEM WORKS



SERVICE JUSTIFICATION



DigitalOcean Managed Load Balancer

- Ensures high availability and distributes traffic

DigitalOcean Droplets (API + Worker)

- Handles request validation and LLM inference (application droplet)

PostgreSQL Database

- Stores users, conversations, and audit data

SERVICE JUSTIFICATION

Google OAuth

- Secure authentication and access control

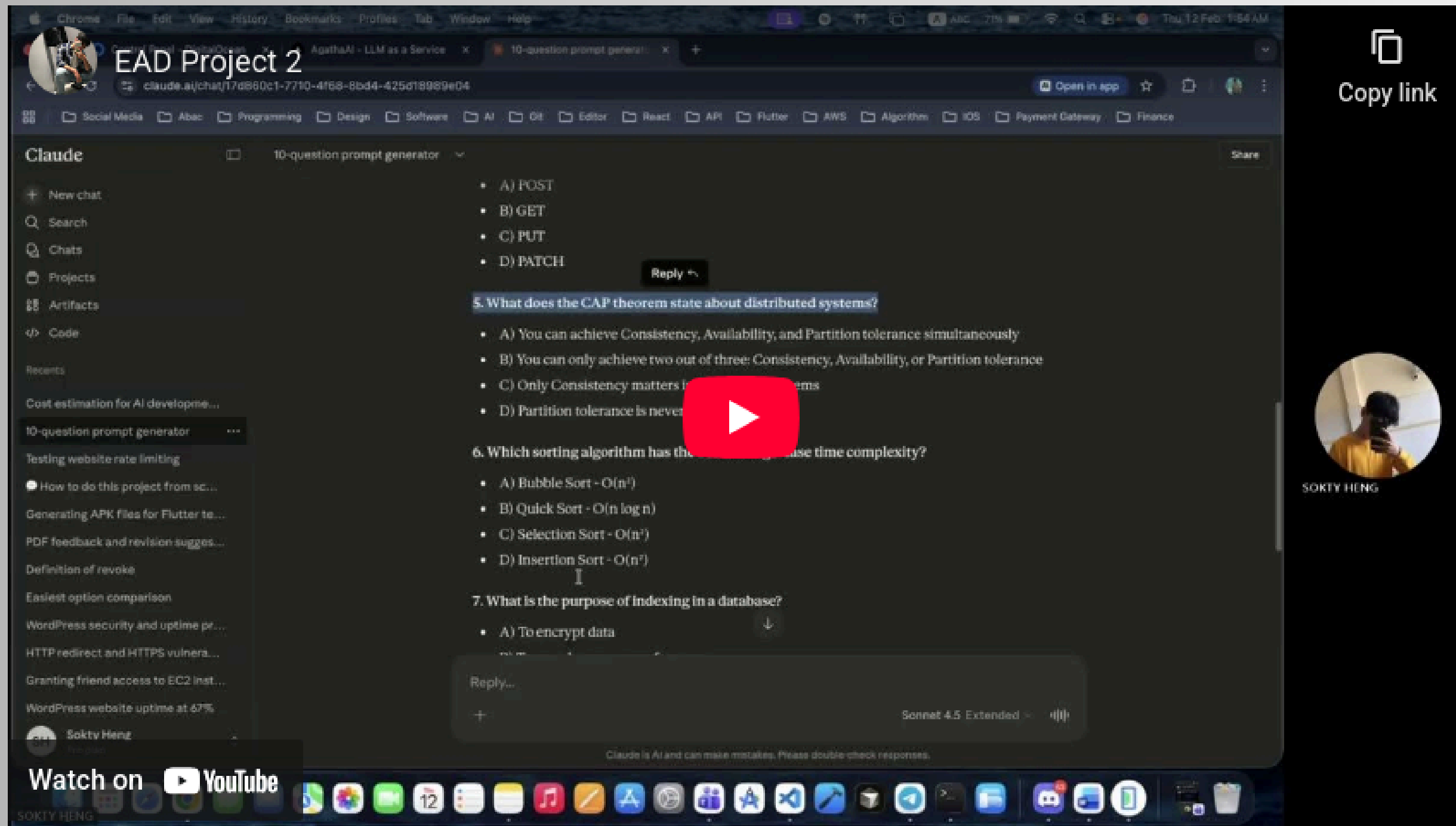
ChatGPT + Gemini APIs

- Balances response quality and cost efficiency

DigitalOcean Monitoring

- Observability and failure detection

DEMO VIDEO



The screenshot displays a web browser window with the address bar showing "claude.ai/chat/17d860c1-7710-4f68-8bd4-425d19989e04". The browser tabs include "EAD Project 2", "AgathaAI - LLM as a Service", and "10-question prompt generator". The browser's bookmark bar lists various categories like Social Media, Abac, Programming, Design, Software, AI, Git, Editor, React, API, Flutter, AWS, Algorithms, iOS, Payment Gateway, and Finance.

The main content area shows a chat interface with the title "Claude" and the subtitle "10-question prompt generator". On the left sidebar, there are options for "New chat", "Search", "Chats", "Projects", "Artifacts", and "Code". Below these are "Recents" and a list of recent prompts, including "Cost estimation for AI developme...", "10-question prompt generator", "Testing website rate limiting", "How to do this project from sc...", "Generating APK files for Flutter te...", "PDF feedback and revision sugges...", "Definition of revoke", "Easiest option comparison", "WordPress security and uptime pr...", "HTTP redirect and HTTPS vulnera...", "Granting friend access to EC2 inst...", and "WordPress website uptime at 67%".

The chat history shows several questions and answers:

- Question 4: "What are the four HTTP methods?"
 - A) POST
 - B) GET
 - C) PUT
 - D) PATCH
- Question 5: "What does the CAP theorem state about distributed systems?"
 - A) You can achieve Consistency, Availability, and Partition tolerance simultaneously
 - B) You can only achieve two out of three: Consistency, Availability, or Partition tolerance
 - C) Only Consistency matters in distributed systems
 - D) Partition tolerance is never achieved
- Question 6: "Which sorting algorithm has the highest worst-case time complexity?"
 - A) Bubble Sort - $O(n^2)$
 - B) Quick Sort - $O(n \log n)$
 - C) Selection Sort - $O(n^2)$
 - D) Insertion Sort - $O(n^2)$
- Question 7: "What is the purpose of indexing in a database?"
 - A) To encrypt data

A red play button is overlaid on the chat content. At the bottom of the chat area, there is a "Reply..." input field and a "Sonnet 4.5 Extended" model selector. A footer note states "Claude is AI and can make mistakes. Please double check responses."

On the right side of the browser window, there is a "Copy link" button and a circular profile picture of a person with the name "SOKTY HENG" below it.

At the bottom of the browser window, there is a dock with various application icons, including Finder, Launchpad, Safari, Mail, Messages, Photos, App Store, and others.

At the bottom left of the image, there is a "Watch on YouTube" button with a YouTube logo and the name "SOKTY HENG" below it.

MONTHLY COST ESTIMATION

Category	Monthly Cost	Note
Load Balancer	\$12/month	DigitalOcean managed LB is HA by default (no second LB required in single region)
Droplet (2x)	\$36 (\$18x2) /month	Configure (2 vCPUs,2 GB,25 GB,3 TB) API + Worker
Database droplet	\$24/month	PostgreSQL
LLM pool	~\$300	Usage based ChatGPT and Gemini
Google Auth	\$0	Free OAuth 2.0
Domain & SSL	~\$16	Domain registration
Monitoring	\$0	Basic: DigitalOcean monitoring (free)

MONTHLY TOTAL COST ESTIMATED

Currency	Amount
USD/month	\$388/month ~ \$400/month
TBH/month (~31.74TBH/USD)	~12,696 THB

TECHNICAL CHALLENGES

- Long-running LLM inference causing blocking requests
- Bursty and unpredictable user traffic
- Cost volatility from usage-based LLM pricing
- Secure handling of sensitive conversational data
- Dependency on third-party LLM providers

**THANK
YOU**

